

ADC Figures of Merit

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ADC Figures of Merit

- Why do we want to establish a figure of merit (FOM)?
 - We want to compare ADCs with different performance metrics
 - We use the FOM to combine several metrics
 - Just like a final course letter grade
- Questions to answer:
 - What are reasonable FOMs?
 - Is there any limit?

FOM₁

$$FOM_1 = f_s \cdot 2^{ENOB}$$

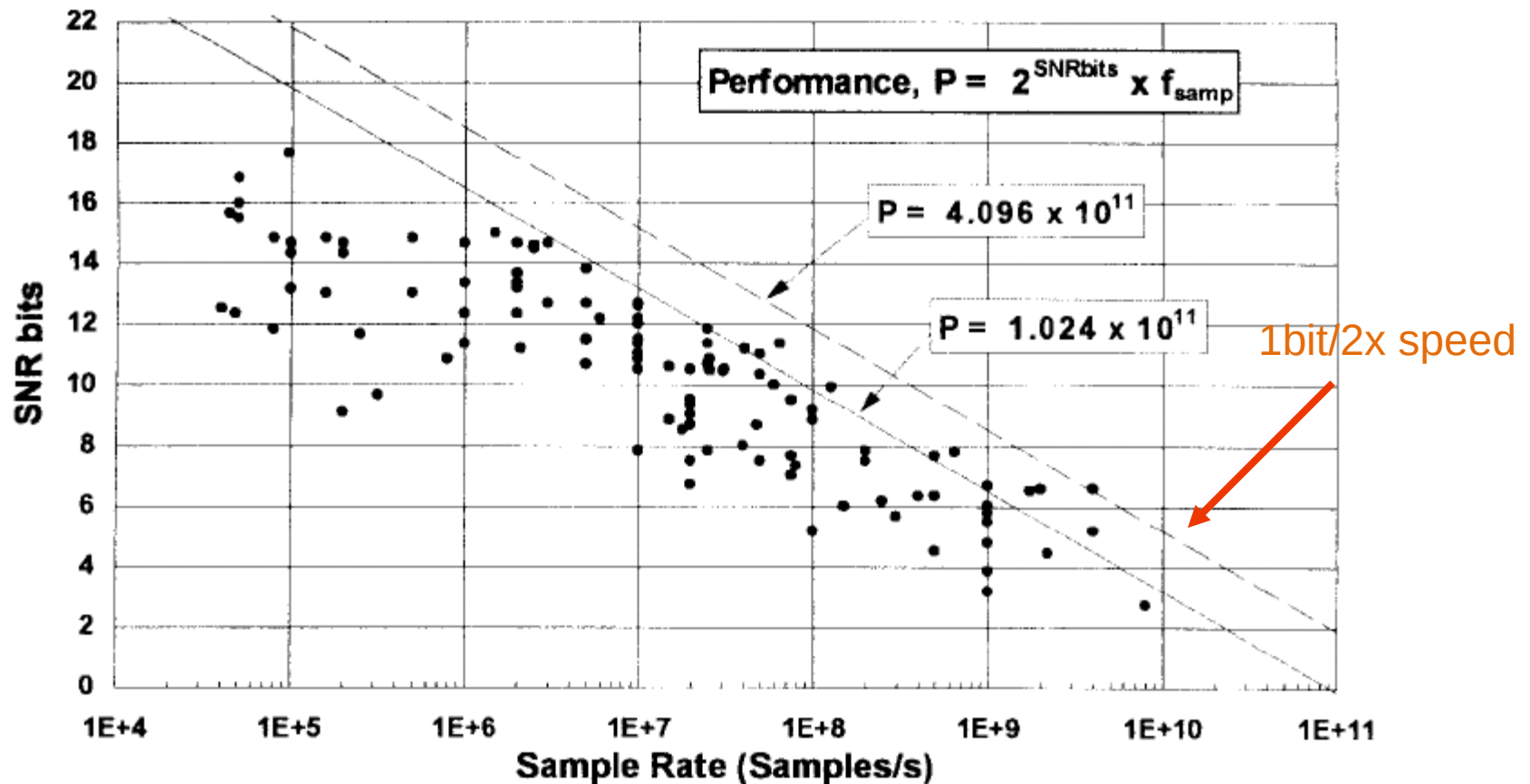
[R. H. Walden, "Analog-to-digital converter survey and analysis," *IEEE Journal on Selected Areas in Communications*, April 1999]

- FOM₁ assumes that 1 extra ENOB = 2x speed
 - True for Nyquist ADCs whose SNR is jitter limited (see Ch. 5):

$$SNR_{jitter} = 20 \log \left[\frac{1}{2\pi \cdot f_{in} \cdot \sigma_t} \right]$$

- Untrue for other ADCs
- Power is not considered in FOM₁

Survey Data by Walden



[Walden, "Analog-to-digital converter survey and analysis," *IEEE J. Selected Areas Comm.*, April 1999]

FOM₂

$$FOM_2 = \frac{\text{Power}}{f_s \cdot 2^{ENOB}}$$

[R. H. Walden, "Analog-to-digital converter survey and analysis," *IEEE Journal on Selected Areas in Communications*, April 1999]

- What it means:
 - How much power does it take for an ADC to convert 1 bit?
- FOM₂ is the **most widely used** ADC FOM
- Typically, circuits power and speed are proportional to each other
 - $f_s \propto g_m/C \propto \text{Power}$
 - FOM₂ captures this tradeoff correctly
- What about power vs. ENOB?
 - 1 extra bit = 2x power?

ENOB vs. Power

- For thermal noise limited ADCs:
 - 1 extra bit $\rightarrow$ 4x less noise power $\rightarrow$ 4x C
 - Power $\propto g_m \propto C \rightarrow$ 1 extra bit = 4x Power
- For mismatch limited ADCs:
 - 1 extra bit $\rightarrow$ 2x comparator #
 - 1 extra bit $\rightarrow$ $\frac{1}{2}$ offset = 4x size = 4x power
 - In total, 1 extra bit = 8x power
- There is no fixed relationship between ENOB and power
- An alternative solution is that
 - We drop ENOB from FOM₂ and only compare ADCs with similar ENOB

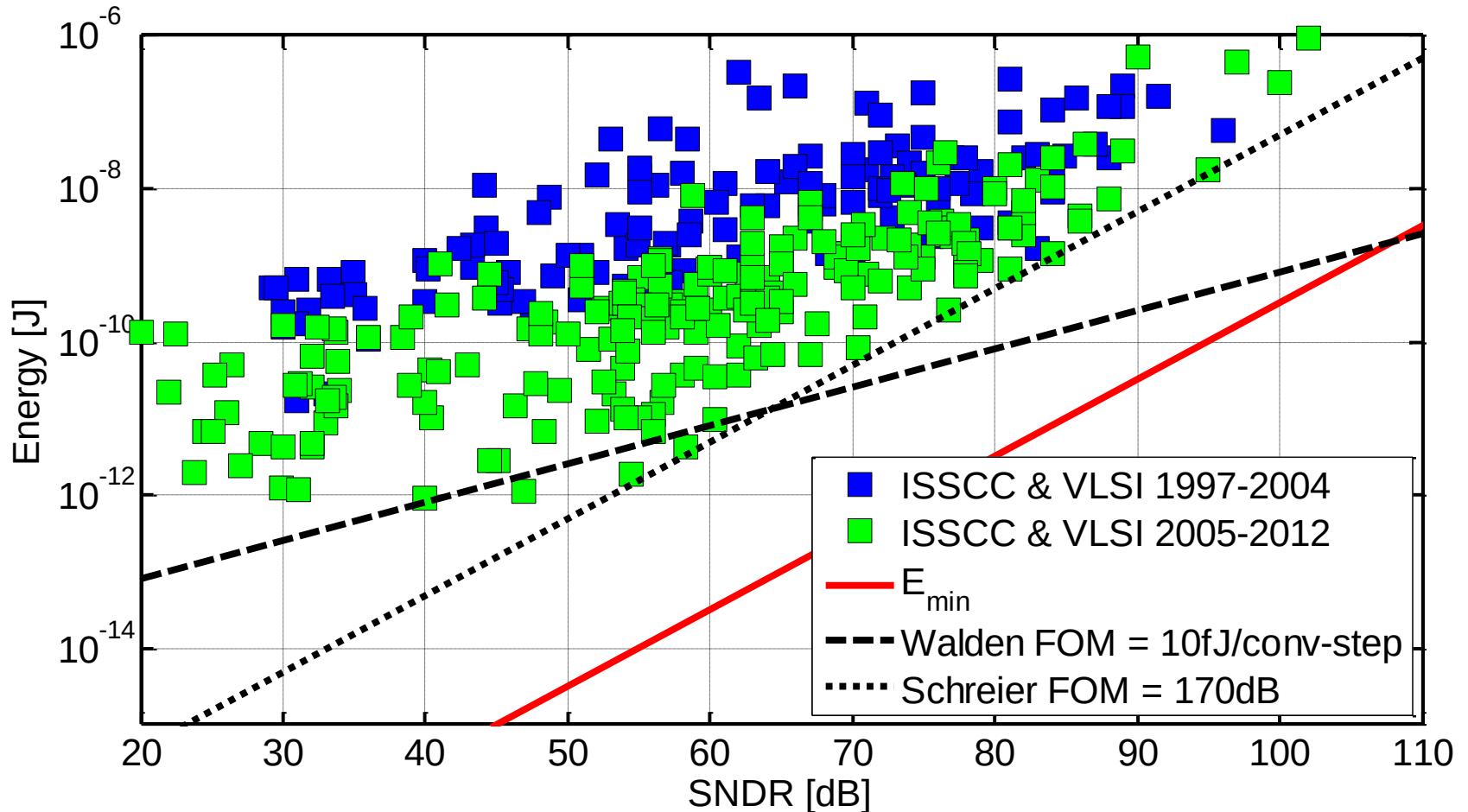
FOM₃

$$FOM_3 = DR(dB) + 10 \log \left(\frac{BW}{P} \right)$$

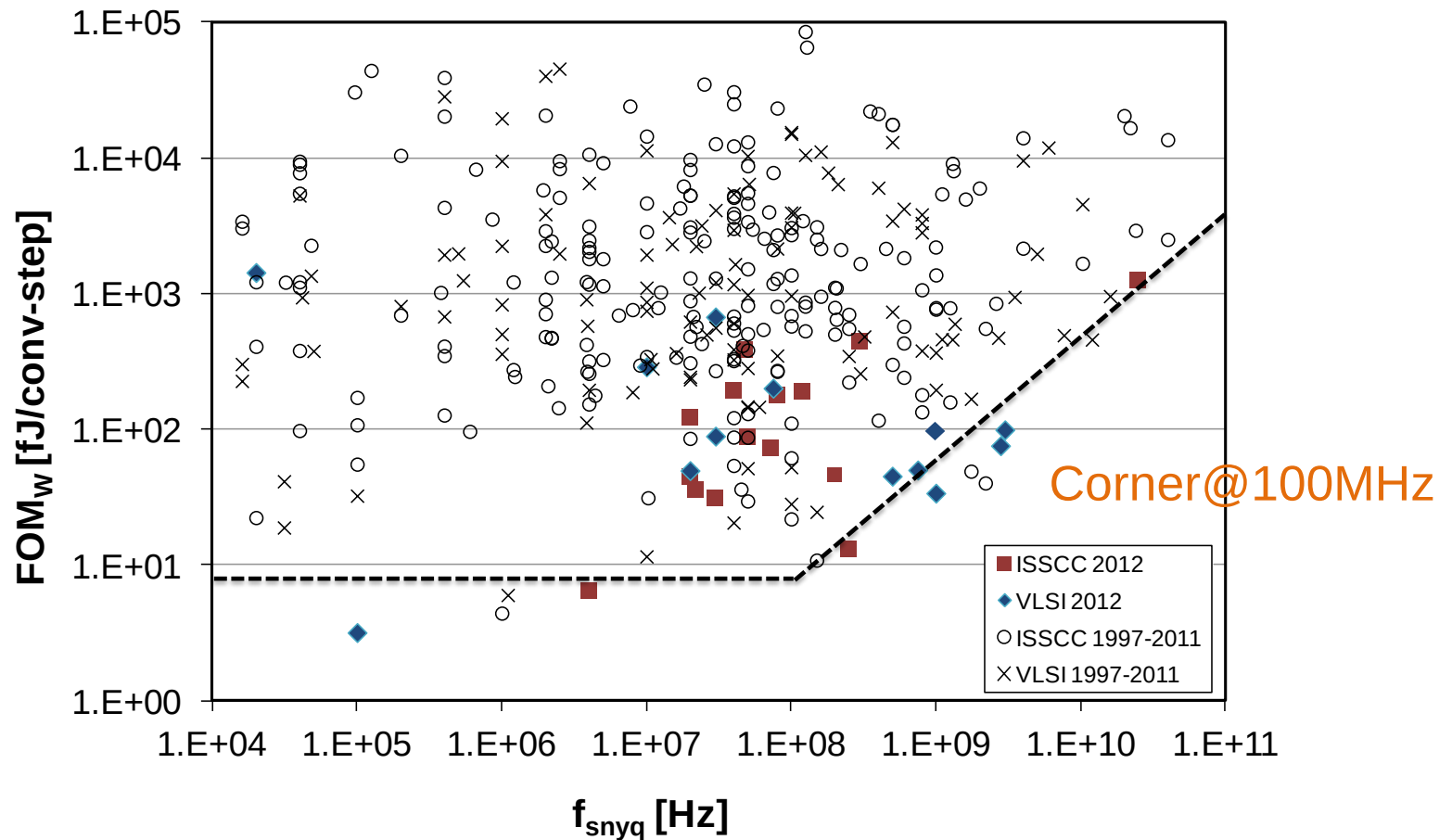
- FOM₃ is a numerical number
 - Sometimes referred as Schreier FOM (Richard Schreier @ ADI)
- Power increases 4x per bit (DR)
 - Correct for thermal noise limited ADCs
- FOM₃ ignores distortion

Recent ADC Survey

From B. Murmann, "Energy limits in A/D converters," IEEE DL talk

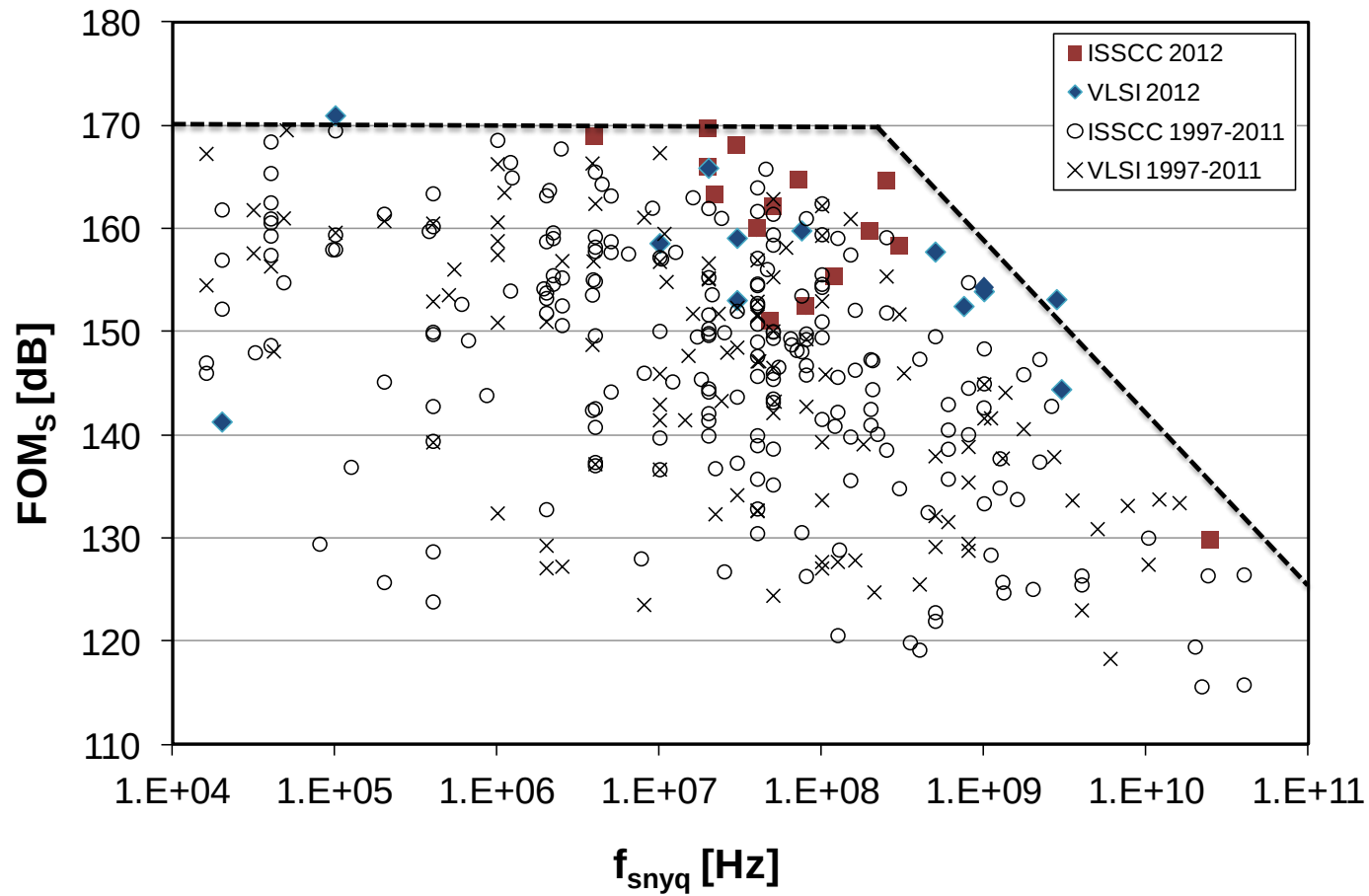


Walden FOM vs. Speed



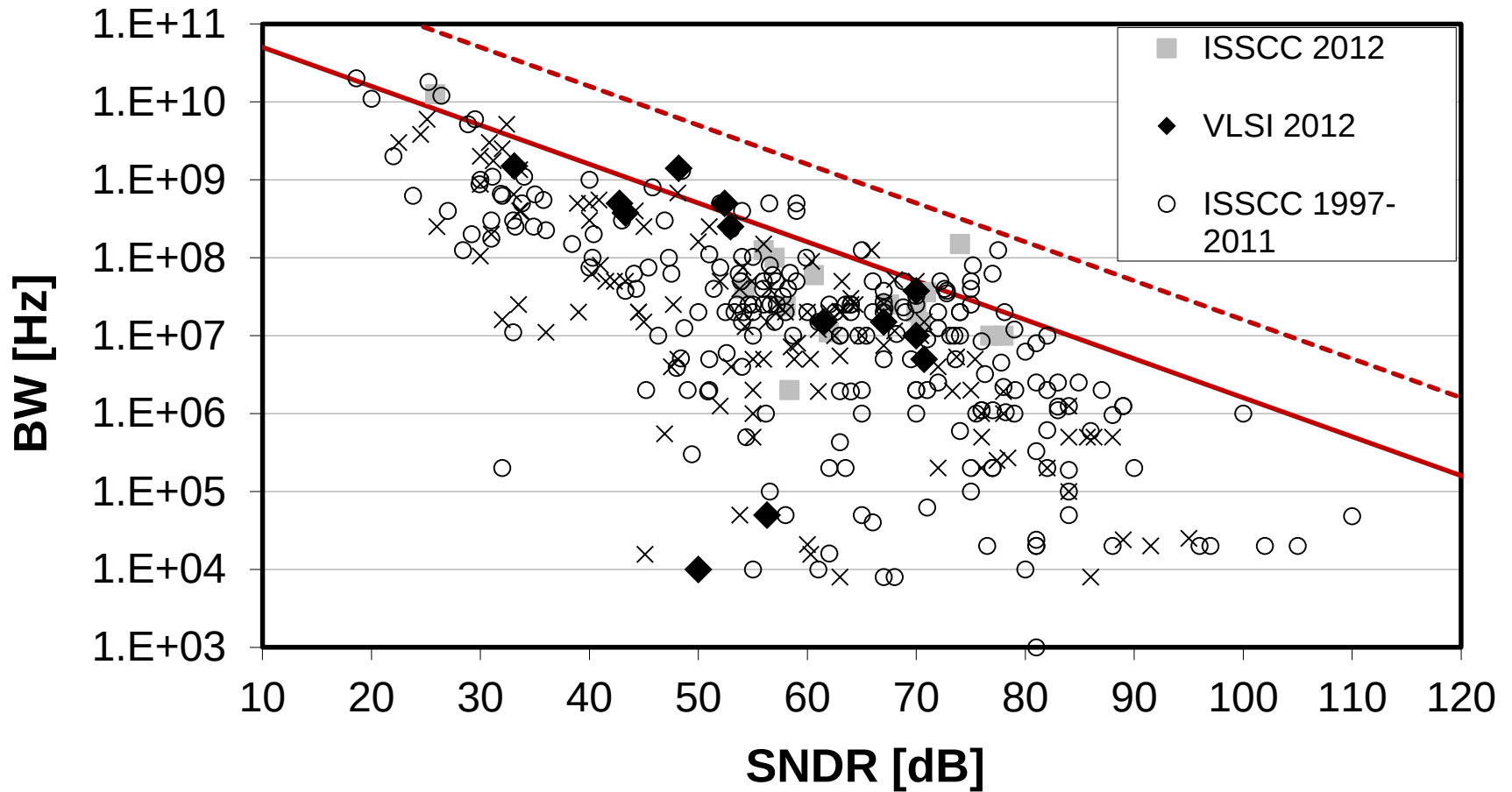
B. Murmann, "ADC Performance Survey 1997-2012," [Online]. Available: <http://www.stanford.edu/~murmman/adcsurvey.html>

Schreier FOM vs. Speed



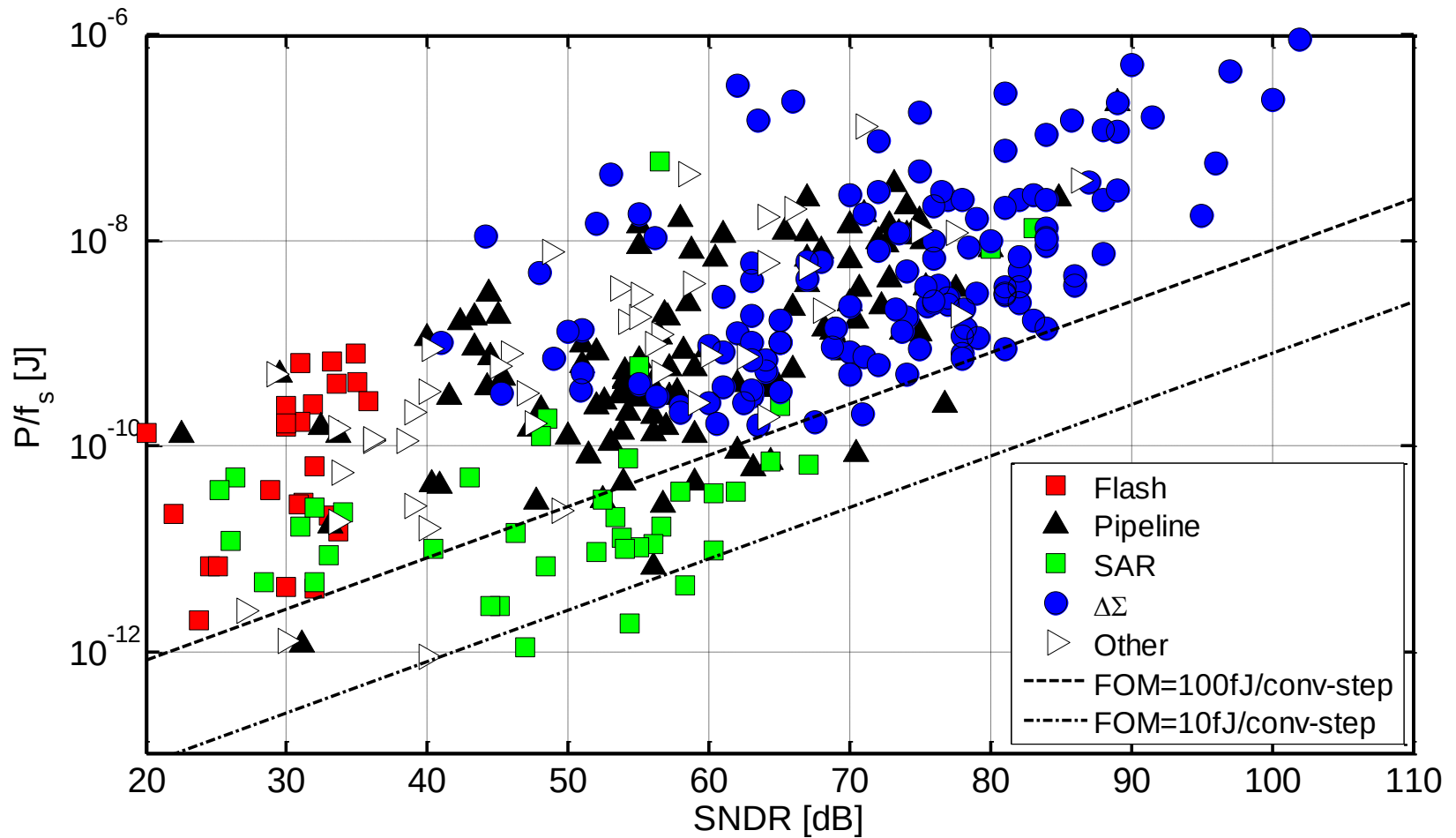
B. Murmann, "ADC Performance Survey 1997-2012," [Online]. Available: <http://www.stanford.edu/~murmman/adcsurvey.html>

SNDR vs. BW



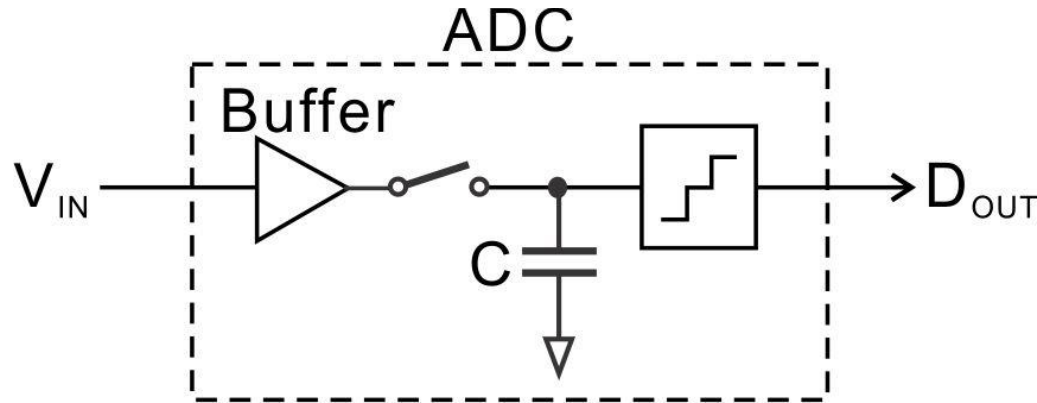
B. Murmann, "ADC Performance Survey 1997-2012," [Online]. Available: <http://www.stanford.edu/~murmman/adcsurvey.html>

Energy by Architecture



B. Murmann, "ADC Performance Survey 1997-2012," [Online]. Available: <http://www.stanford.edu/~murmman/adcsurvey.html>

ADC Power Limit

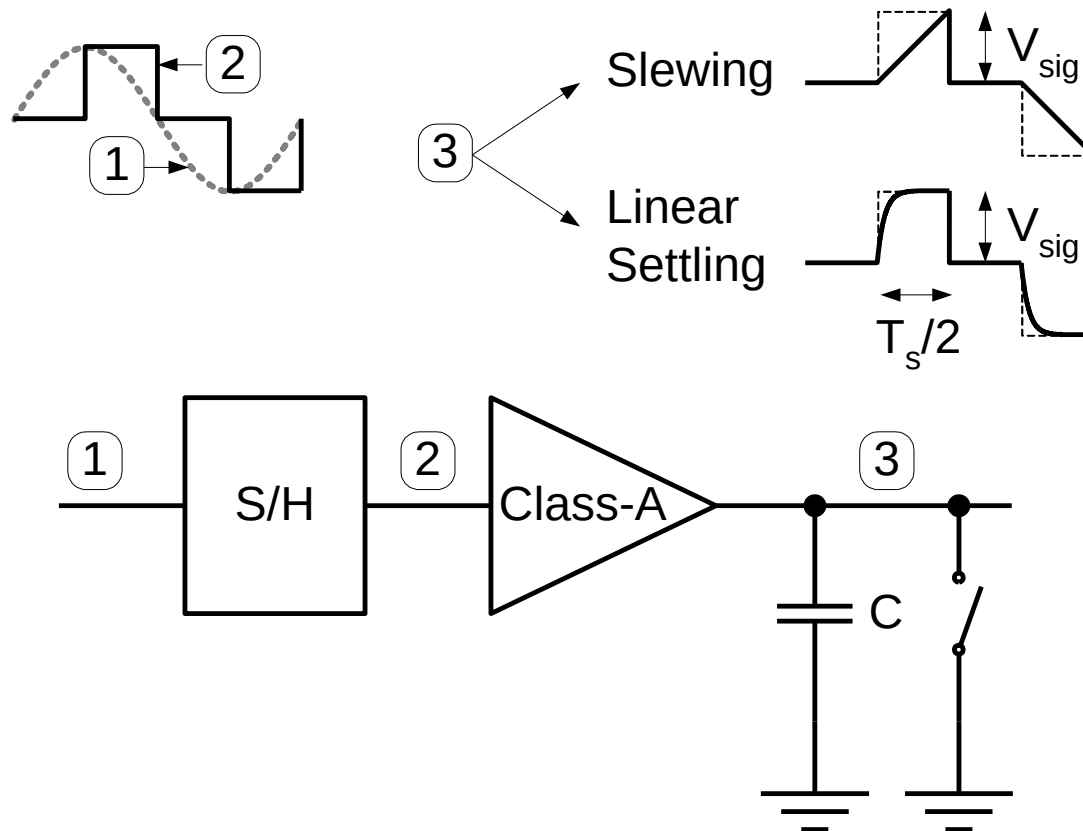


- Let us assume that an ADC can be considered as an ideal buffer with 100% efficiency and a zero-power zero-noise quantizer, then:

$$P = \frac{1}{2} C V_{DD}^2 f_s \quad V_n^2 = \frac{kT}{C} \quad SNR = \frac{\frac{1}{2} \left(\frac{V_{DD}}{2} \right)^2}{V_n^2} = \frac{V_{DD}^2 / 8}{kT / C} = \frac{C V_{DD}^2 / 8}{kT}$$

$$P = 4kT \cdot SNR \cdot f_s$$

Switched Capacitor Circuits



Case 1: 100% Slewing

$$I_{bias} = C \cdot \frac{dV}{dt} = C \cdot \frac{V_{sig}}{T_s / 2} = 4 \cdot C \cdot V_{sig} \cdot f_{sig}$$

$$P = 2 \cdot V_{sig} \cdot I_{bias} \qquad SNR = \frac{0.5 \times V_{sig}^2}{k_B T / C}$$

$$\boxed{\therefore P = 16 k_B T \cdot SNR \cdot f_{sig}}$$

Case 2: 100% Linear Settling

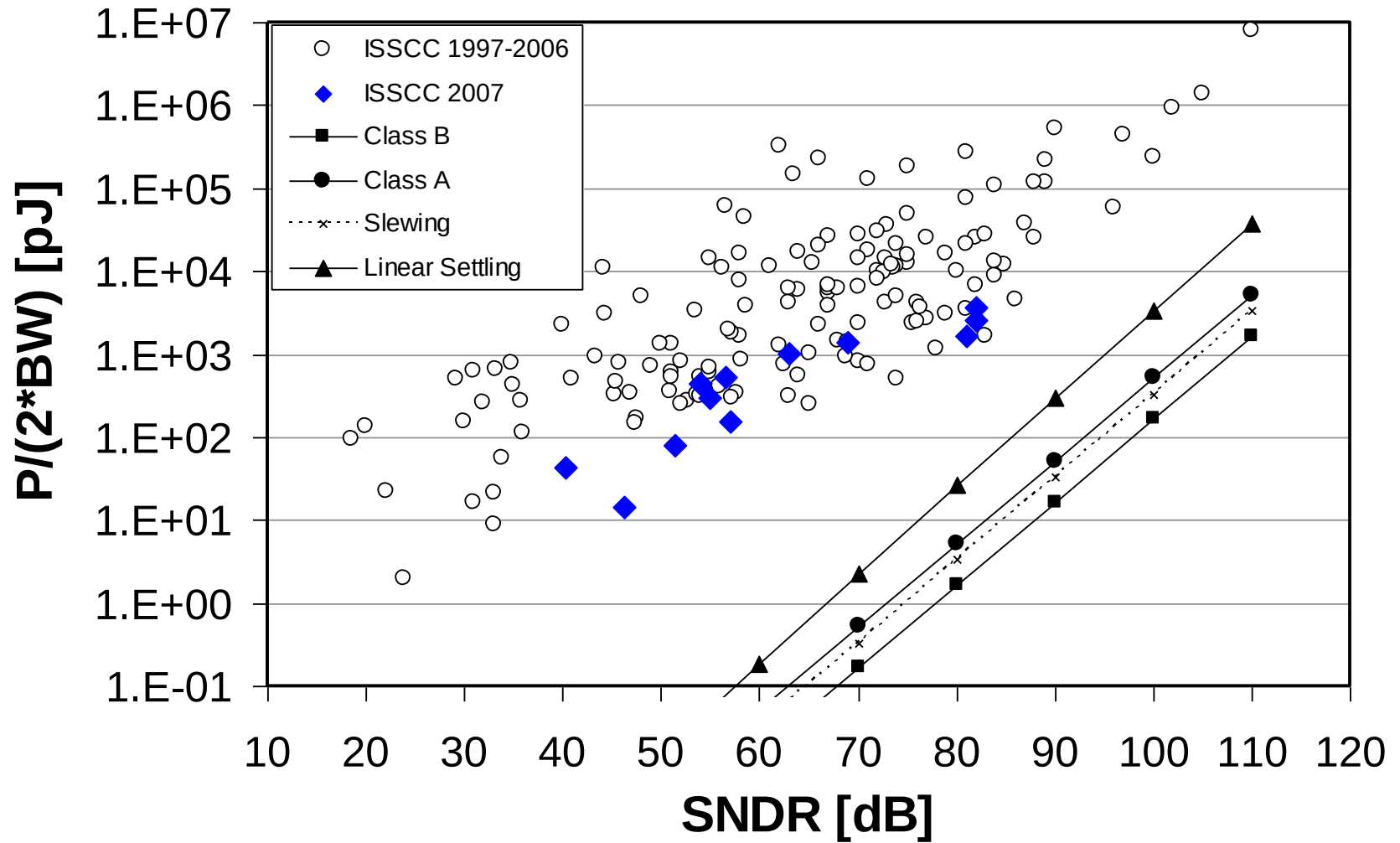
$$I_{bias} = C \cdot \left. \frac{dV}{dt} \right|_{max} = C \cdot \left. \frac{d}{dt} \right|_{max} \left[V_{sig} \left(1 - e^{-t/\tau} \right) \right] = C \cdot \frac{V_{sig}}{\tau}$$

Number of settling time constants: $N = \frac{T_s / 2}{\tau}$

$$\therefore P = 16 \cdot N \cdot k_B T \cdot SNR \cdot f_{sig}$$

- Much worse
 - E.g. N=6.9 for settling to 0.1% precision

Limit Lines



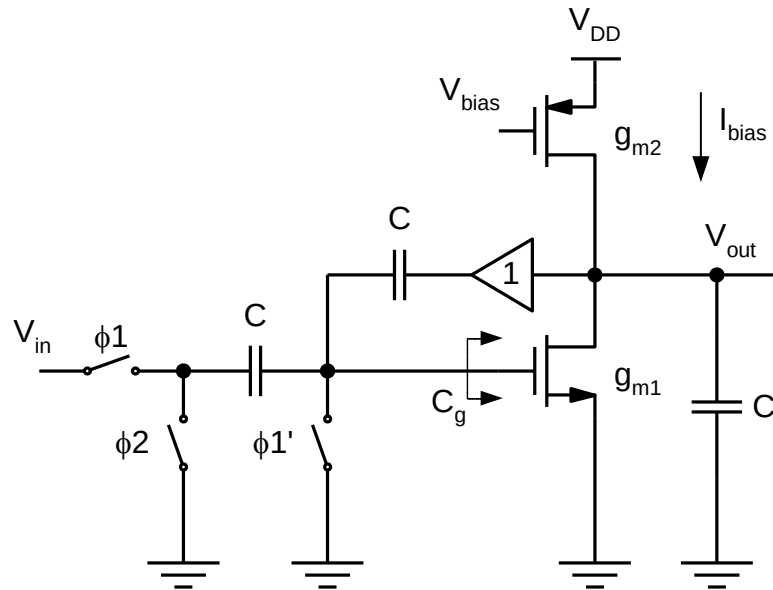
Discussion

- Still orders of magnitude away from limits
 - What contributes to these large gaps?
 - Must keep in mind that ADCs are not just a single capacitor circuit
- Slope of limit lines is much steeper than fit to experimental data
- The following analysis factors in practical considerations
 - Not fundamental, but somewhat unavoidable in today's implementations

Design Space Partitioning

- High SNR
 - Complexity ~ 1 (e.g. first integrator in sigma-delta ADC)
 - Limited by thermal noise
- Moderate SNR
 - Complexity $\sim \text{Bits}$ (e.g. pipelined ADC)
 - Partly limited by thermal noise
- Low SNR
 - Complexity $\sim 2^{\text{Bits}}$ (e.g. flash ADC)
 - Limited by matching, quantization noise

High SNR SC-Stage (1)



- Considerations
 - Noise is multiple of $k_B T / C$ (n_f)
 - Swing is only a fraction of V_{DD} (α)
 - Feedback factor (β)
 - g_m / I_D is upper bounded if slewing must be avoided

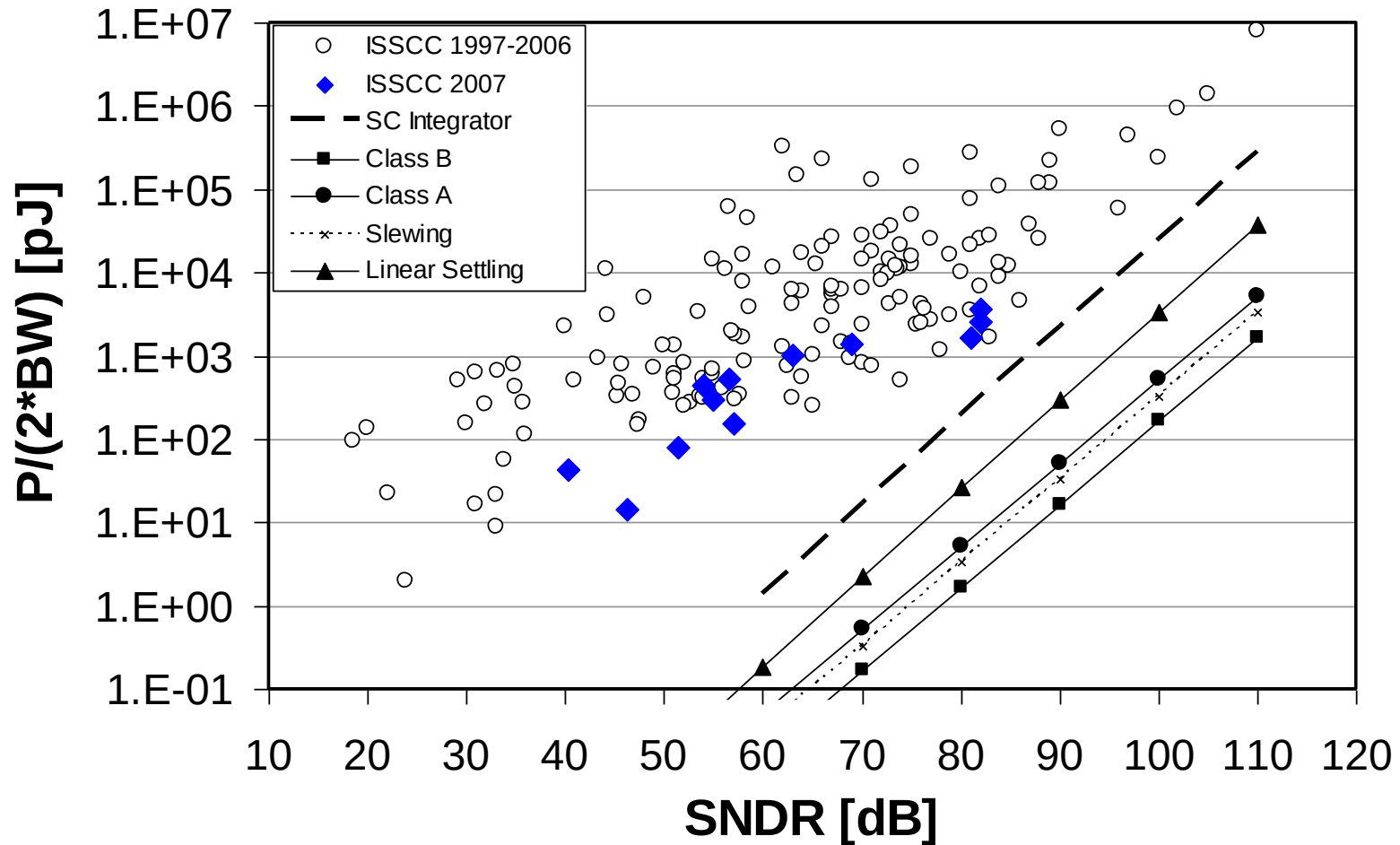
High SNR SC-Stage (2)

To avoid slewing: $\frac{g_{m1}}{I_{bias}} \leq \frac{1}{\beta \cdot V_{sig}}$

$$\therefore P = 16 \cdot N \cdot n_f \cdot \frac{1}{\alpha} \cdot k_B T \cdot SNR \cdot f_{sig} \cdot \max \left(1, \frac{1}{\frac{g_{m1}}{I_{bias}} \cdot \beta \cdot V_{sig}} \right)$$

- Graph on following slide shows result assuming
 - $n_f=5$, $\alpha=2/3$, $\beta=0.5$, onset of slewing

High SNR SC-Stage (3)

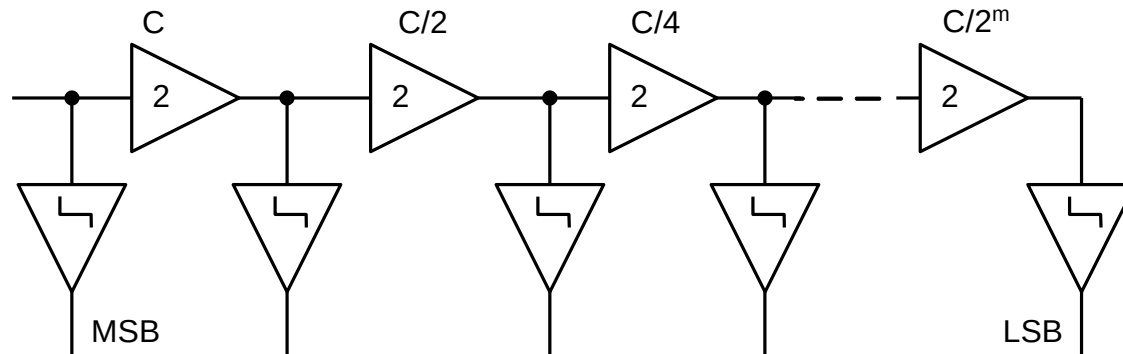


- Close to experimental data at high SNDR!

Medium SNR

- Pipeline ADC using SC stages
 - Partially limited by thermal noise

Pipeline ADC



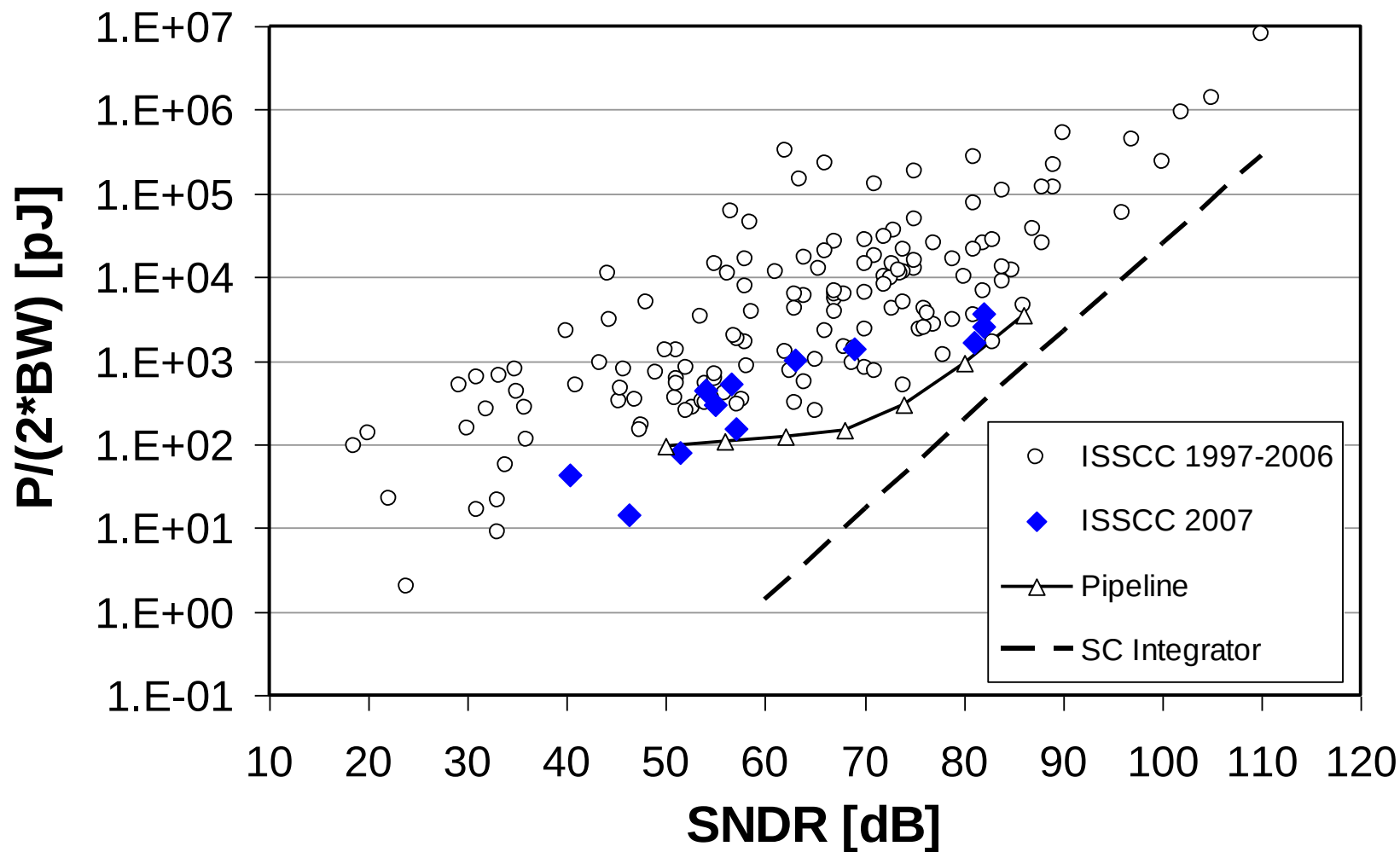
- Theoretical near optimum power scaling
 - Scale capacitance by gain of preceding stage
 - Stage 1 consumes half of total power
 - Adding one bit means power goes up 4x
- Caveat
 - Usually impractical to scale capacitors down to $C/2^m$

Stage Scaling Example

Number of Amplifiers	13	12	11	10
Stage Capacitances	1	1/4	1/16	1/64
	1/2	1/8	1/32	1/128
	1/4	1/16	1/64	1/128
	1/8	1/32	1/128	1/128
	1/16	1/64	1/128	1/128
	1/32	1/128	1/128	1/128
	1/64	1/128	1/128	1/128
	1/128	1/128	1/128	
	1/128	1/128		
	1/128			
ΣC	2.03	0.54	0.17	0.086
C_{single}	1/2	1/8	1/32	1/128
Relative Power Pipeline/Single SC Stage ($\Sigma C/C_{\text{single}}$)	4.06	4.32	5.44	11.01

- Example is simplistic, but in line with state-of-the art
 - 10bits $\sim 0.5\text{mW/MSample/s}$, 12bits $\sim 2\text{mW/MSample/s}$

Pipeline ADC Limit Line



Low SNR

- Power of matching limited class-B circuit [Kinget, CICC 1996]

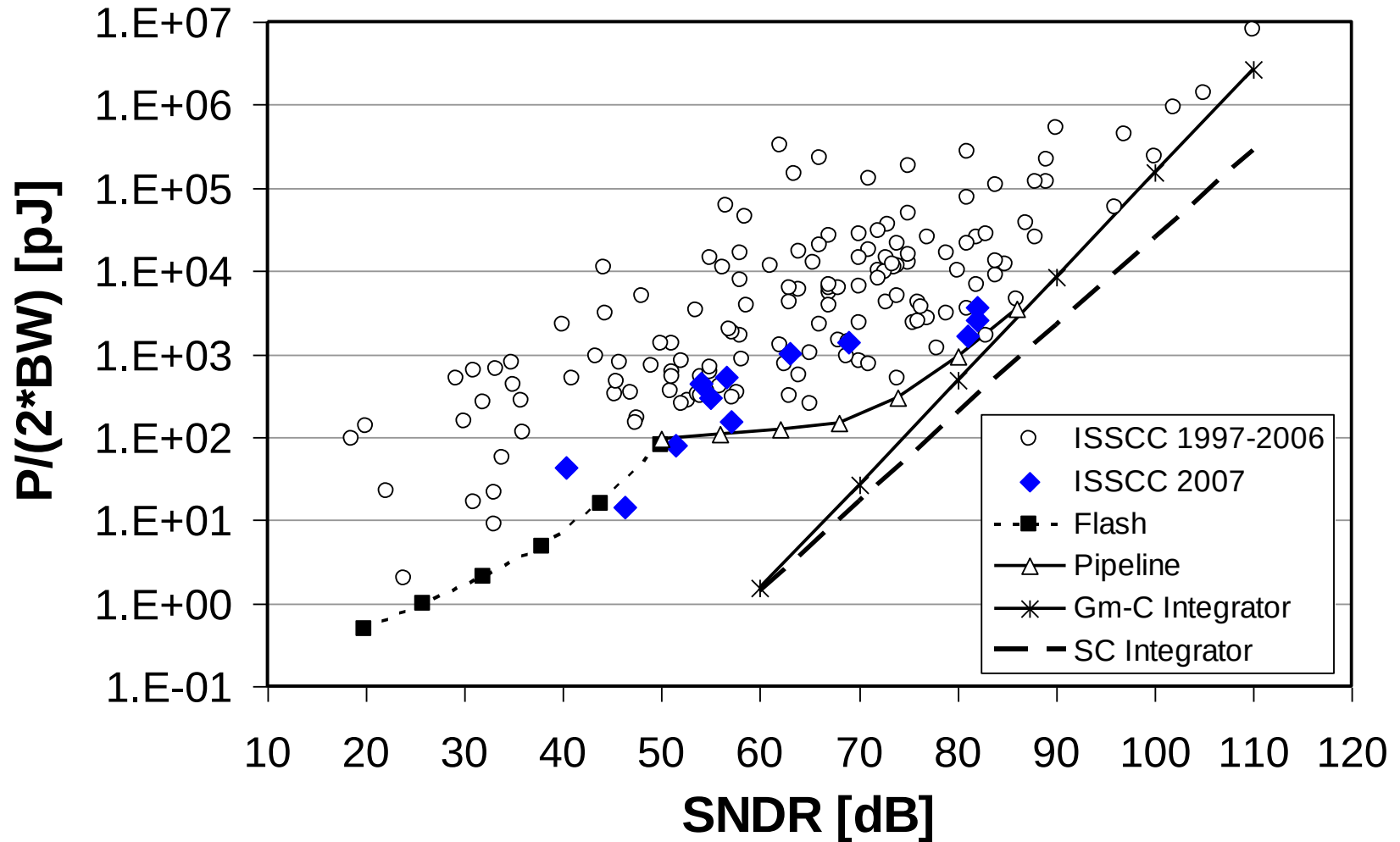
$$P = 24 \cdot C_{ox} \cdot A_{VT}^2 \cdot f_{sig} \cdot \left(\frac{V_{sig,rms}}{3 \cdot \sigma_{Vos}} \right)^2$$

- Refined result for flash ADC, assuming
 - Class-A, 1/2 LSB matching with 3σ -confidence, 2^B components, additional E_{dyn} per clock cycle, partial supply usage (α)

$$P = \left(12\pi \cdot \frac{1}{\alpha} \cdot C_{ox} \cdot A_{VT}^2 \cdot 2^{3B} + 2 \cdot E_{dyn} \cdot 2^B \right) \cdot f_{sig}$$

- Example: $\alpha=2/3$, $C_{ox}=15\text{fF}/\mu\text{m}^2$, $A_{vt}=3\text{mV}\cdot\mu\text{m}$, $E_{dyn}=60\text{fJ}$ (~ 10 gates in $0.13\mu\text{m}$ CMOS)

End Result

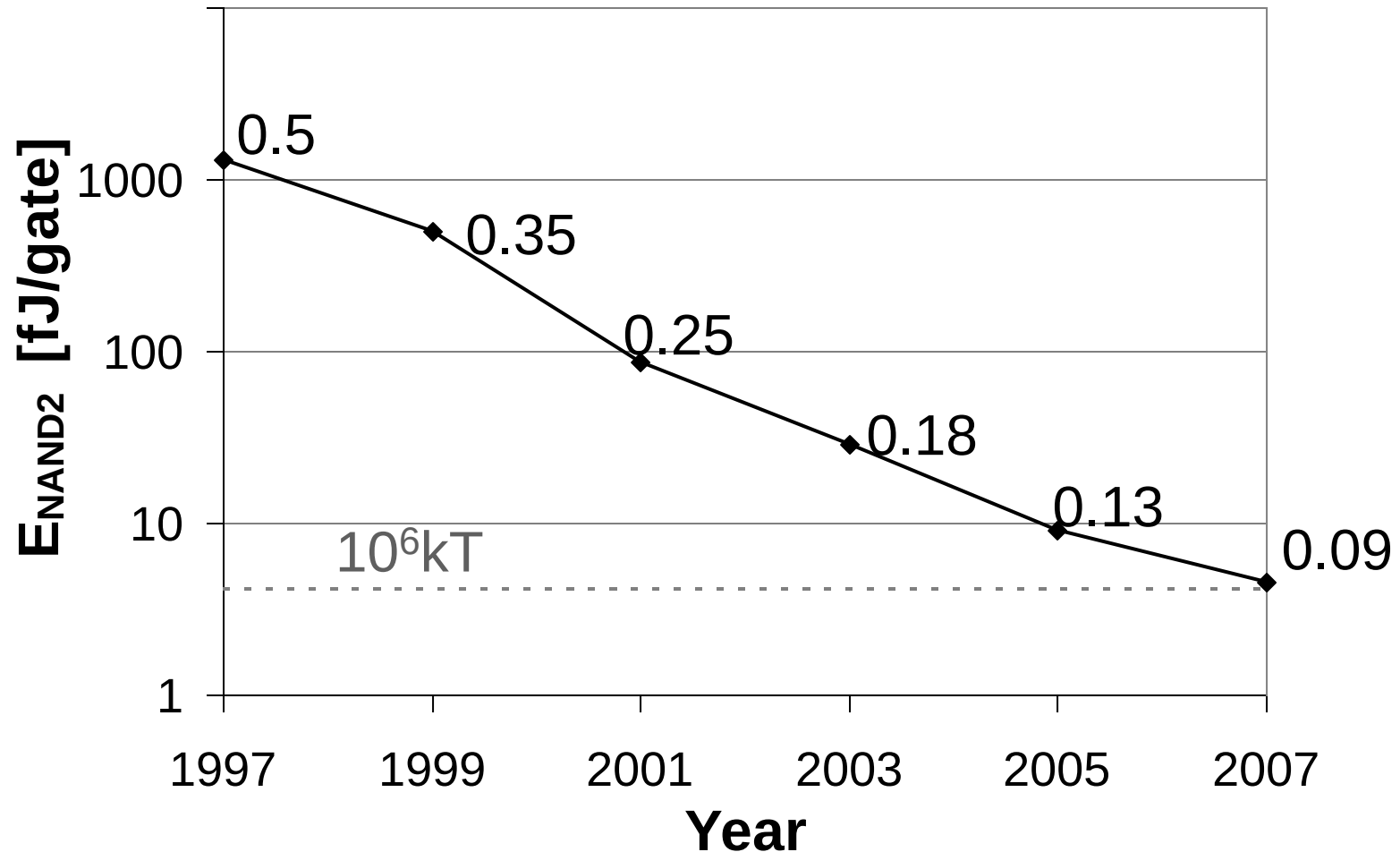


CMOS Scaling

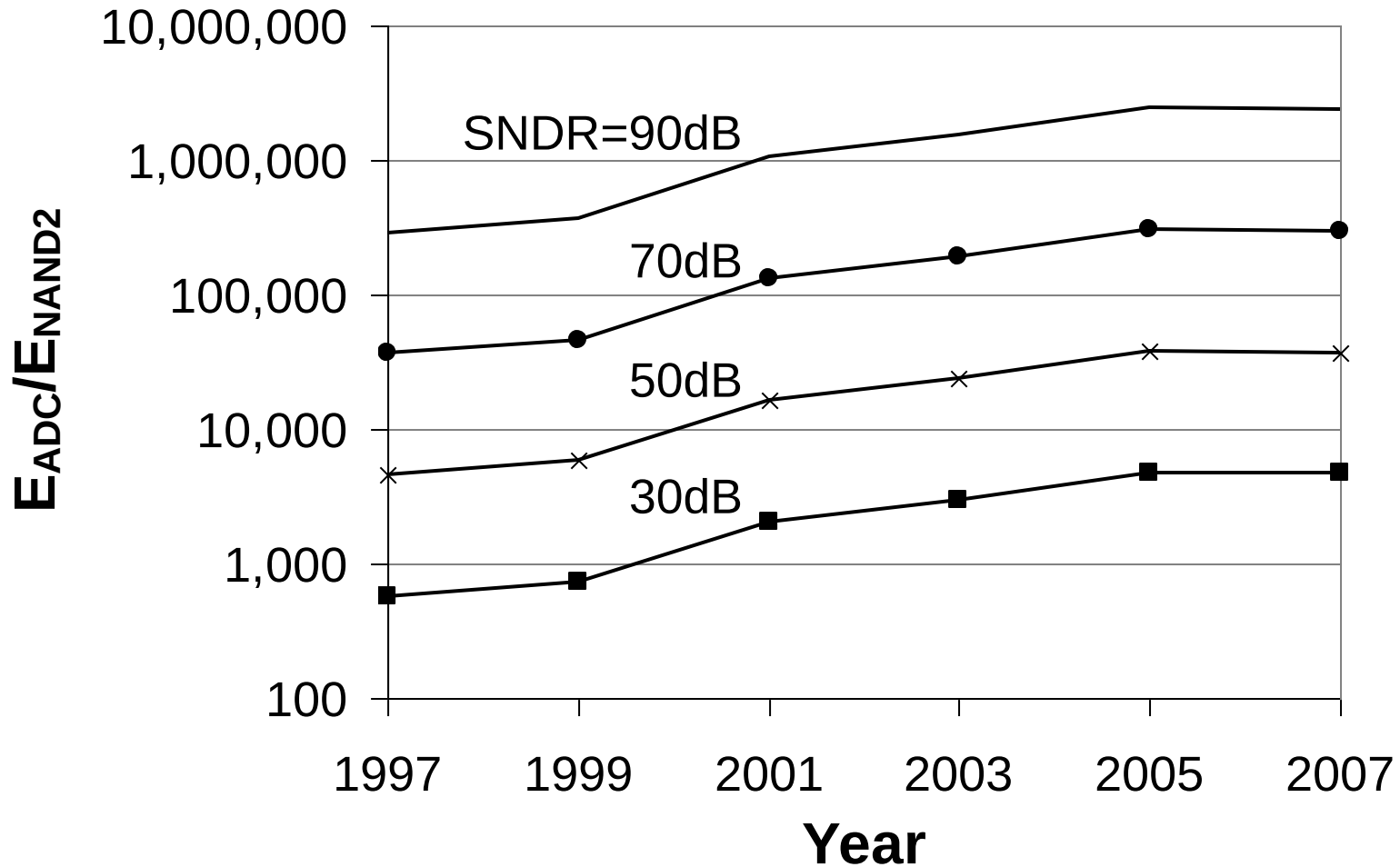
- CMOS scaling brings mixed news
 - Good news: Lower dynamic power, faster speed, higher g_m/I_D given the same speed, ...
 - Bad news: Lower V_{DD} , lower transistor intrinsic gain, ...
- New technologies brings new opportunities
 - Vacuum tubes → Bipolar → CMOS → FINFET, nanowire, graphene
- What should we do?
 - Invent new topologies that take advantage of CMOS scaling and obviate its drawbacks
 - Digitally assisted analog circuit design
 - Minimalistic analog circuits
 - Digital compensation of resulting non-idealities
 - Time/phase domain analog signal processing

Digital Logic Energy Trend

Mainstream ADC technologies, standard logic library data



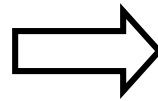
ADC/Digital Logic Energy Ratio



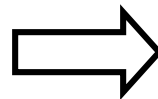
Energy Ratio in 2007

- Interpretation for digitally enhanced ADCs (energy centric)

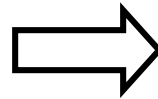
SNDR	$E_{\text{ADC}}/E_{\text{NAND2}}$
30	4,679
50	37,432
70	299,479
90	2,396,045



Additional digital processing is costly!



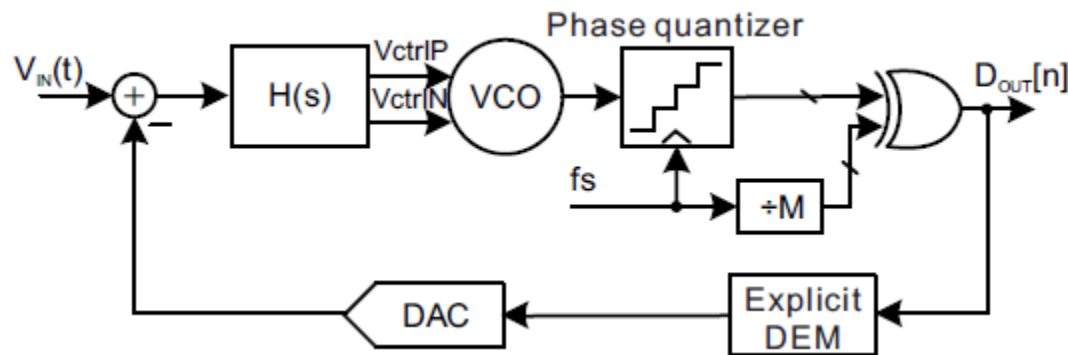
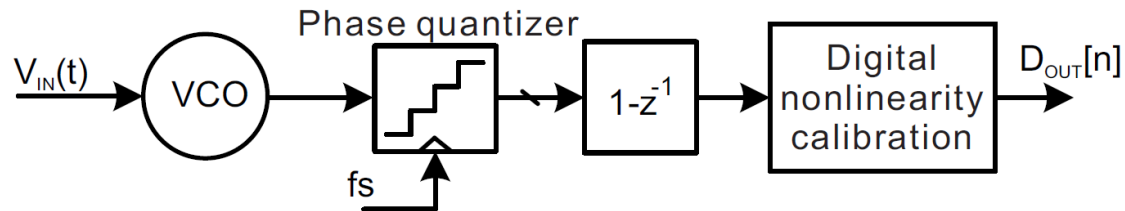
Several tens of thousand gates are "free"



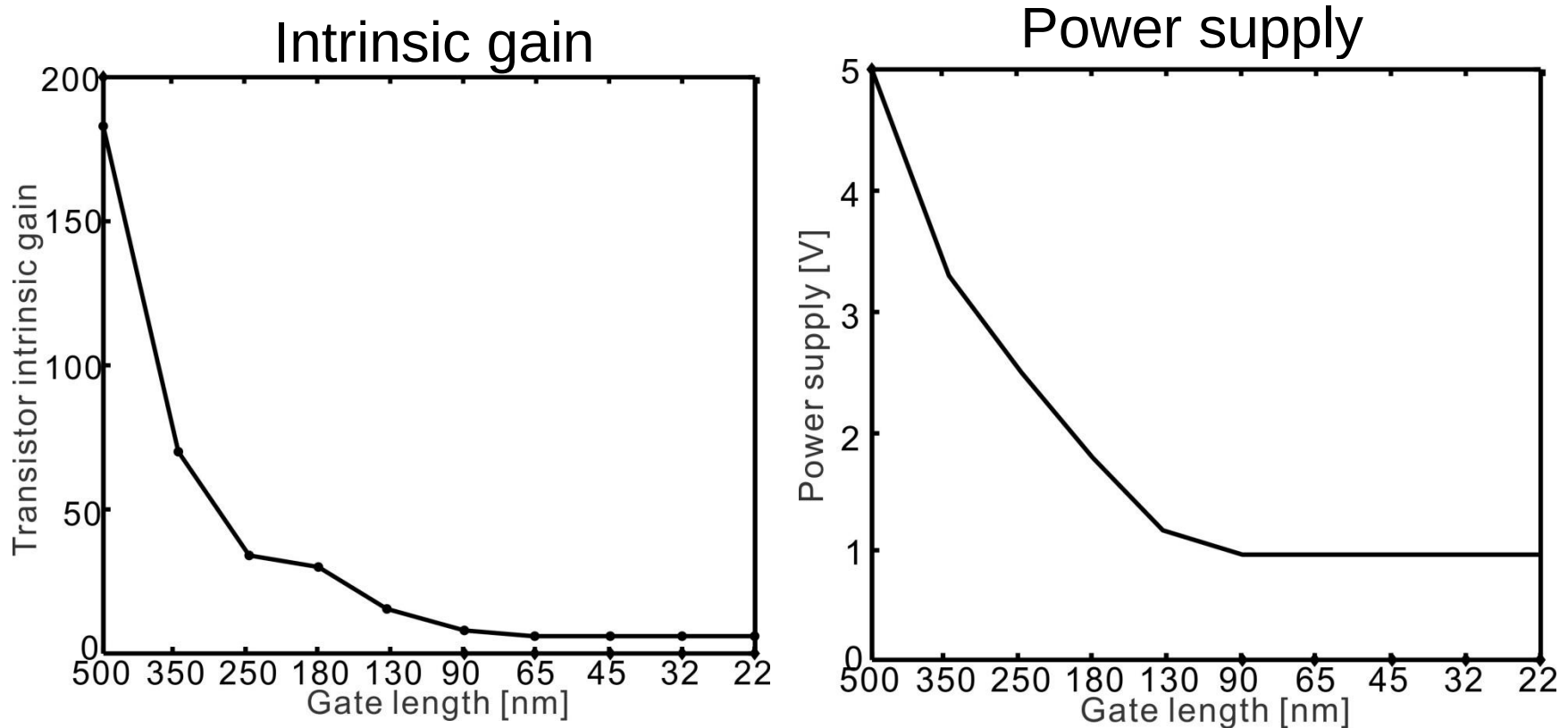
Use as many gates as you can fit...

Time-Domain ADC

- Using ring voltage-controlled oscillator (VCO)
 - VCO can be use as both quantizer and integrator

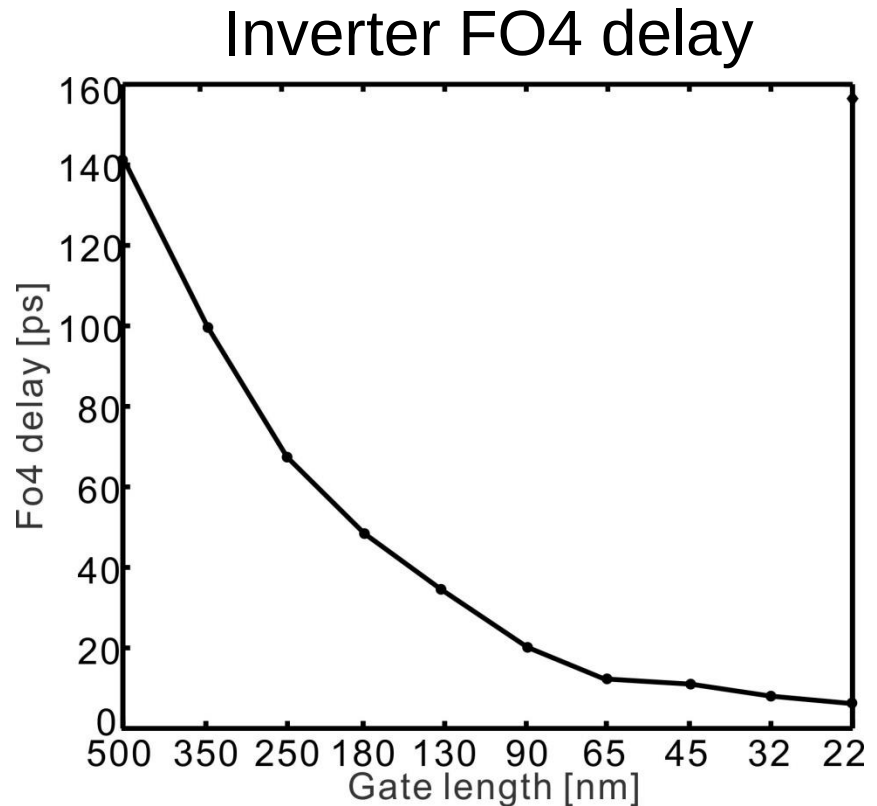
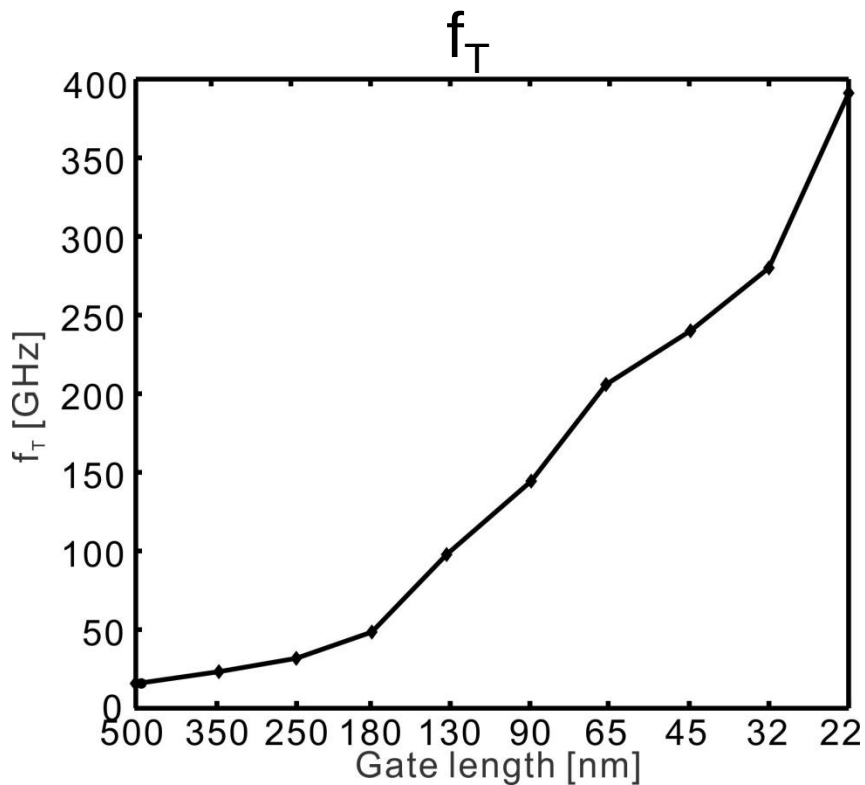


Technology scaling trend (1)



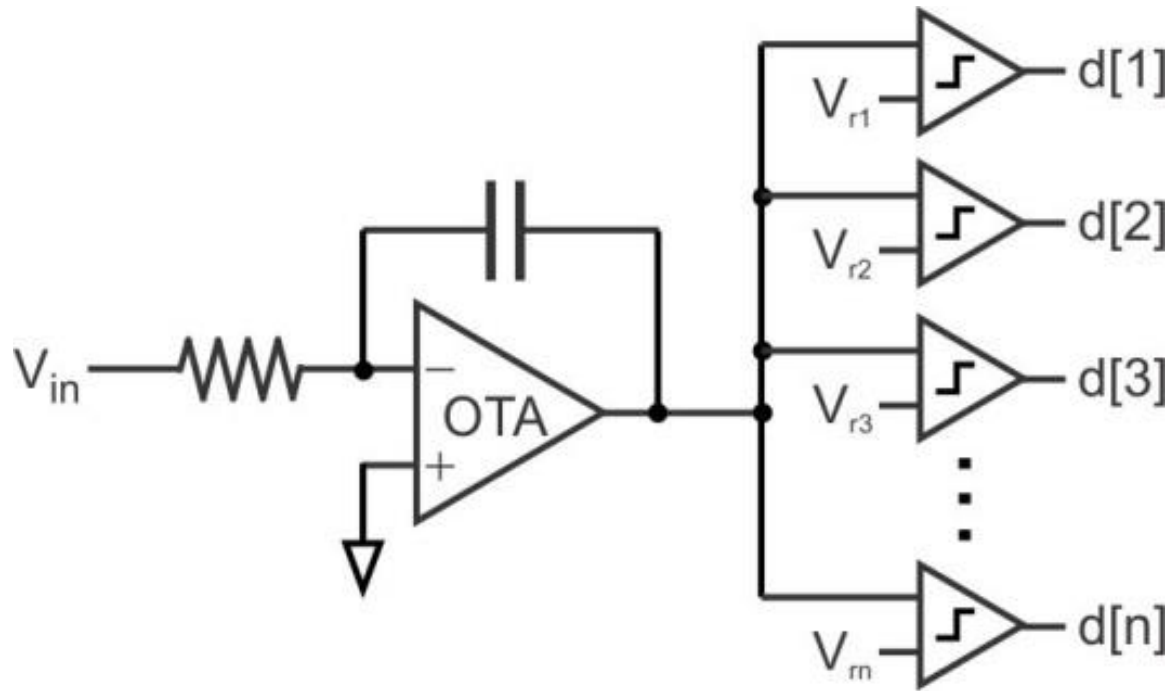
Conventional analog signal processing (ASP) schemes are not scaling friendly!

Technology scaling trend (2)



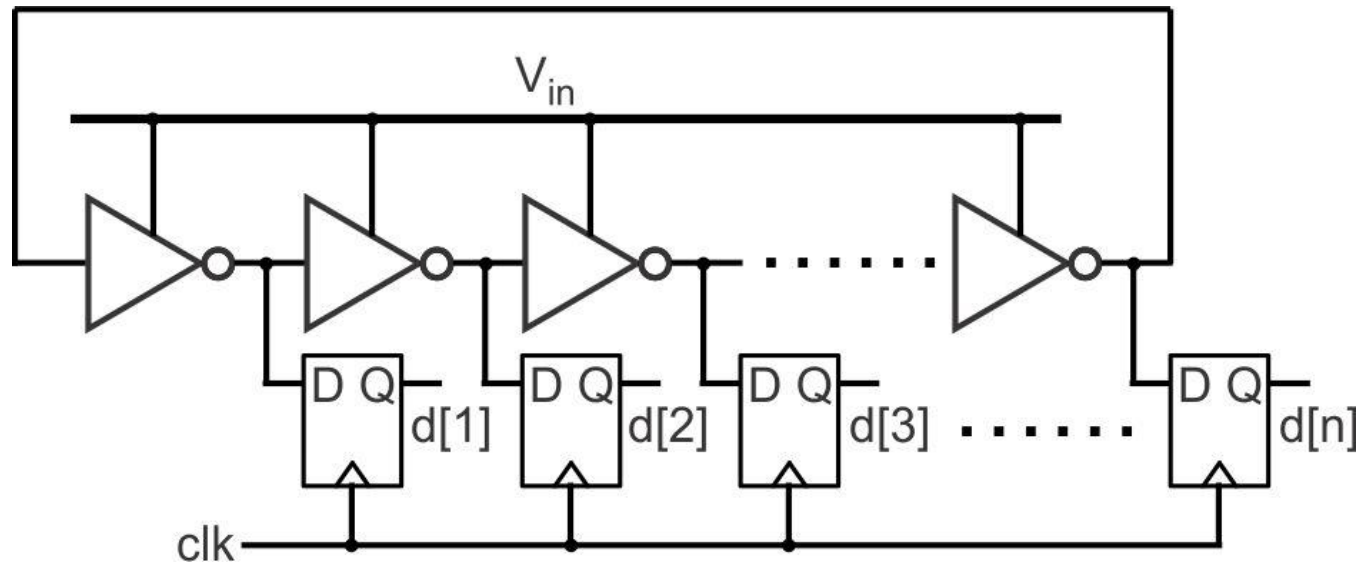
It is desirable to develop scaling-friendly time-domain ASP techniques.

Conventional integrator and quantizer



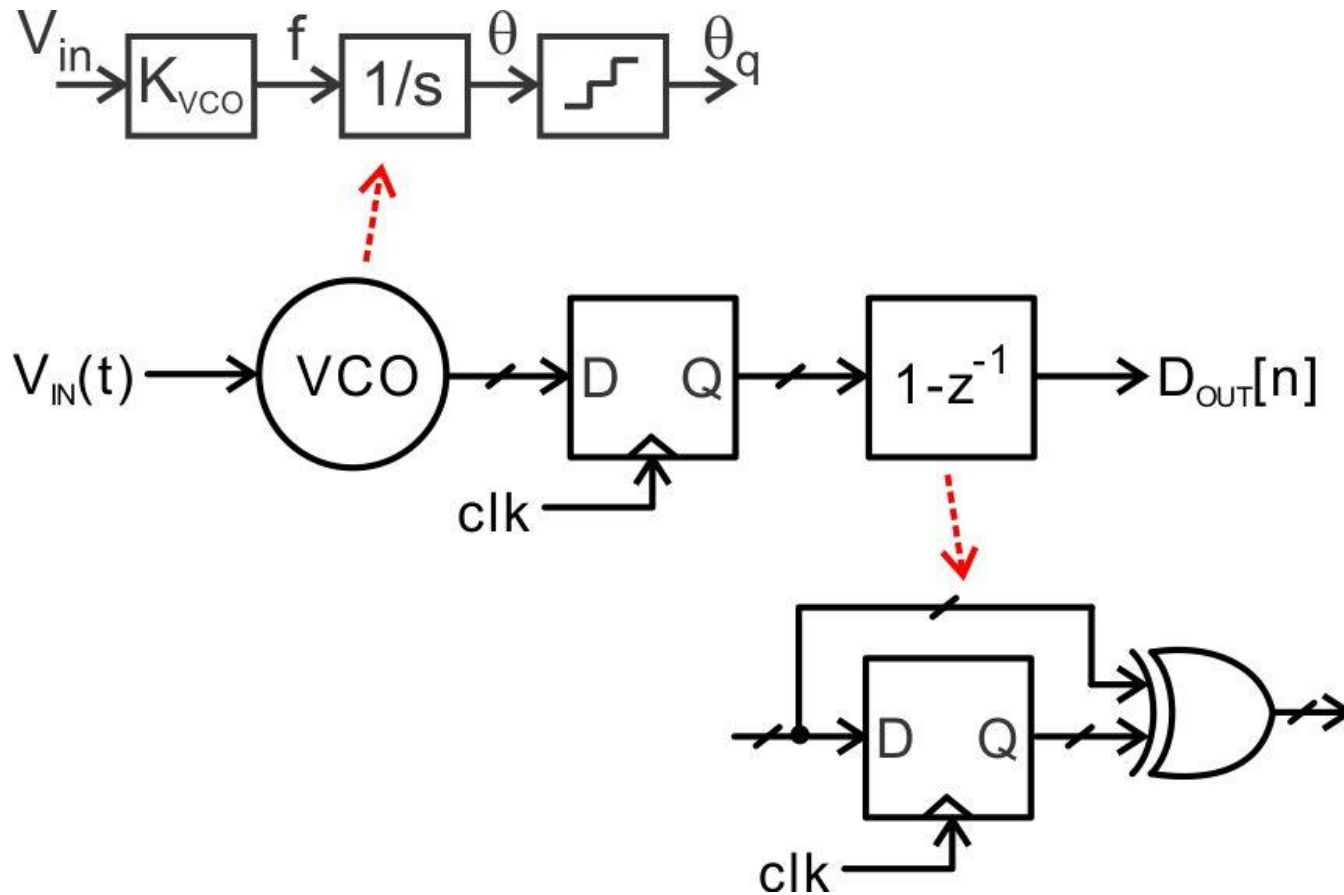
- Need high-gain OTA ☹
- Scaling unfriendly ☹
- Require large power supply ☹
- Need precision comparators ☹
- Large area/power ☹

VCO integrator and quantizer

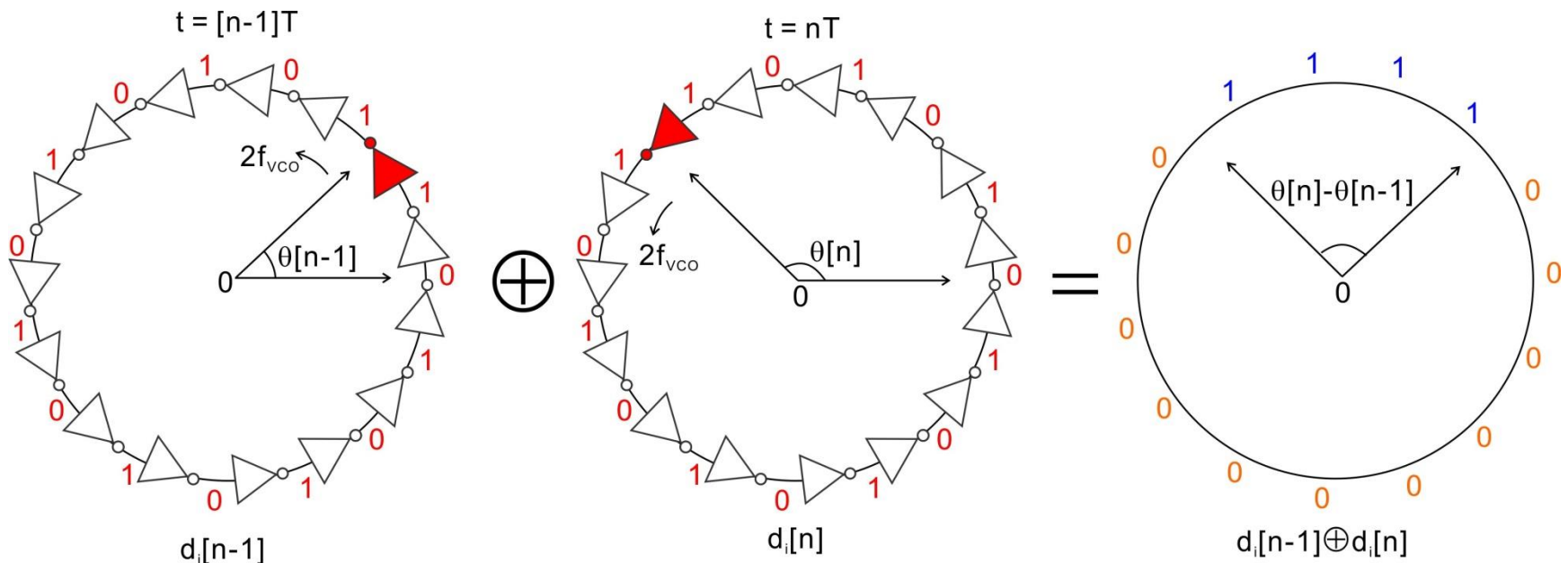


- Intrinsic integration ☺
- Scaling friendly ☺
- Work under low supply ☺
- Easy to design ☺
- Intrinsic multi-bit quantizer ☺
- No need for precise comparator ☺
- Small area/power ☺

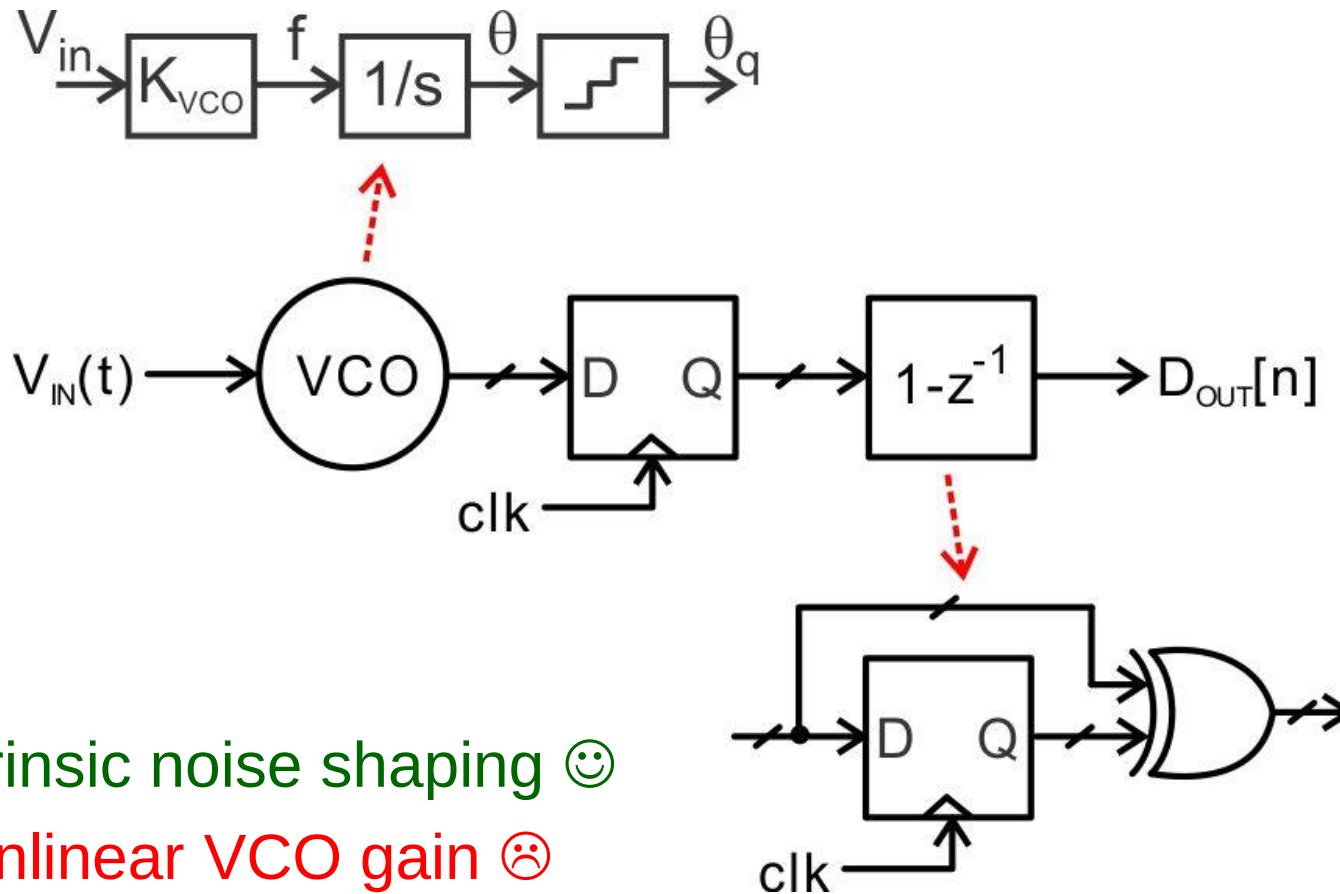
VCO-based $\Delta\Sigma$ ADC



Geographical view

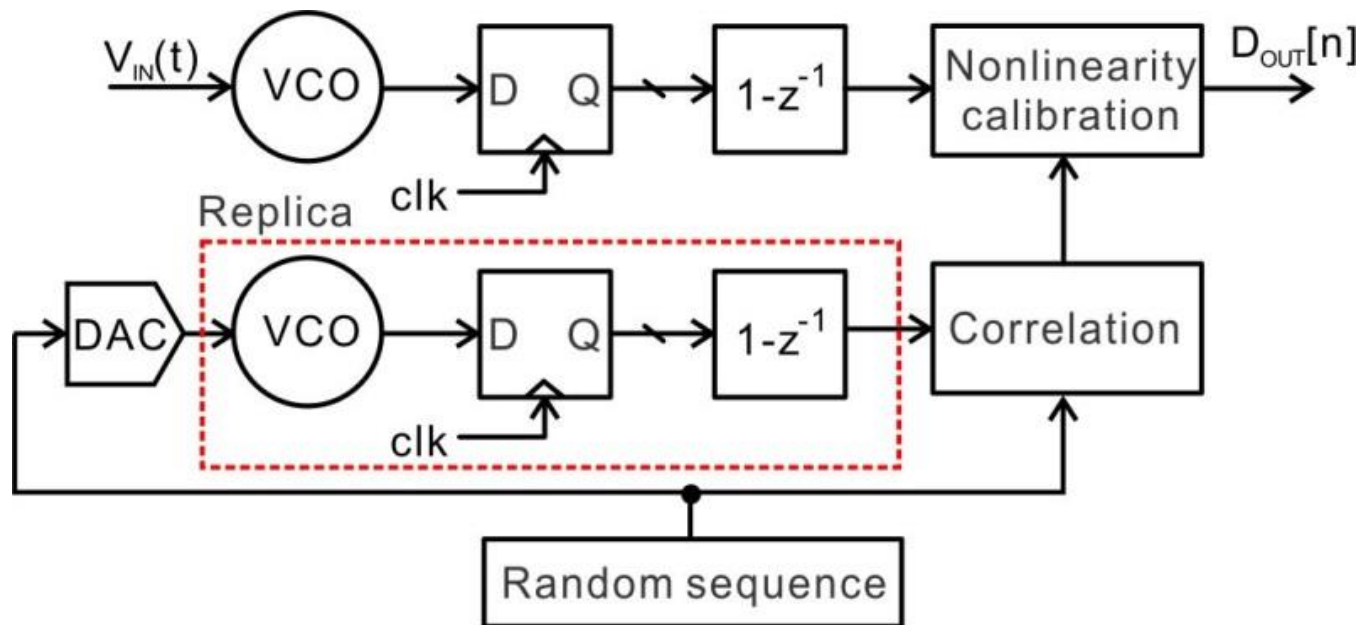


VCO-based $\Delta\Sigma$ ADC



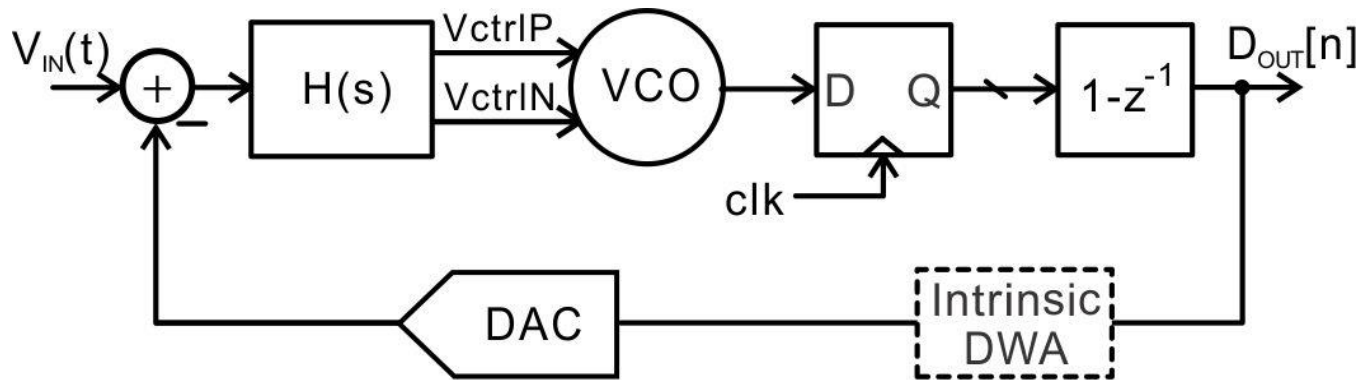
- Intrinsic noise shaping ☺
- Nonlinear VCO gain ☹

VCO-based ADC review (1)



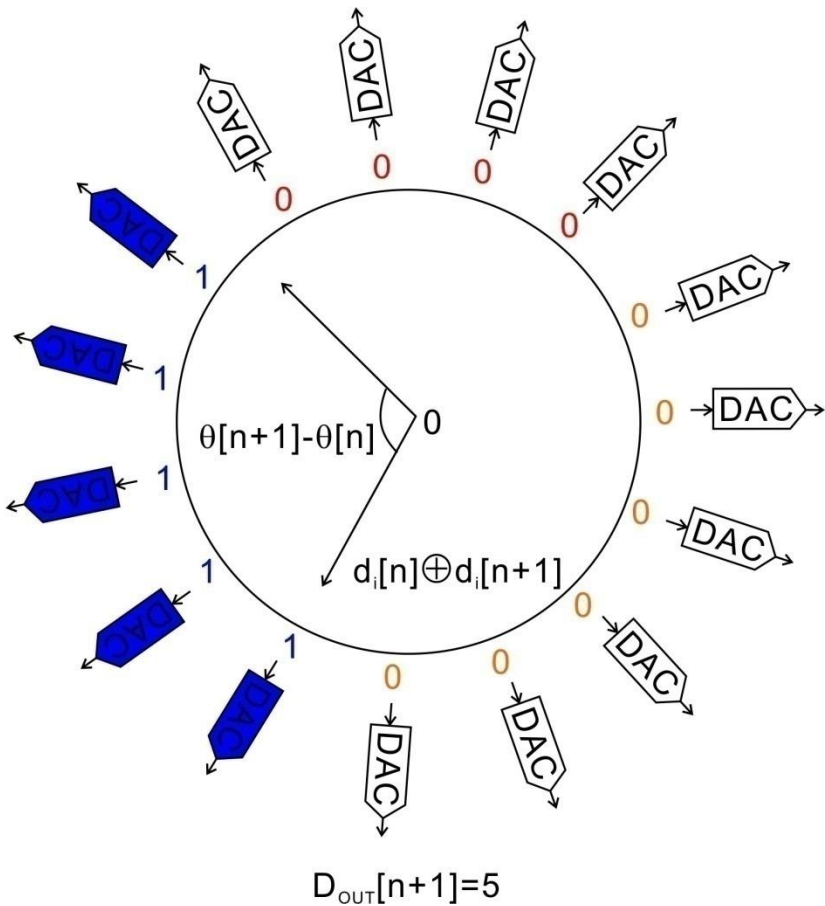
- Need complicated calibration algorithm, require extra dithering DAC, and rely on replica matching

VCO-based ADC review (2)

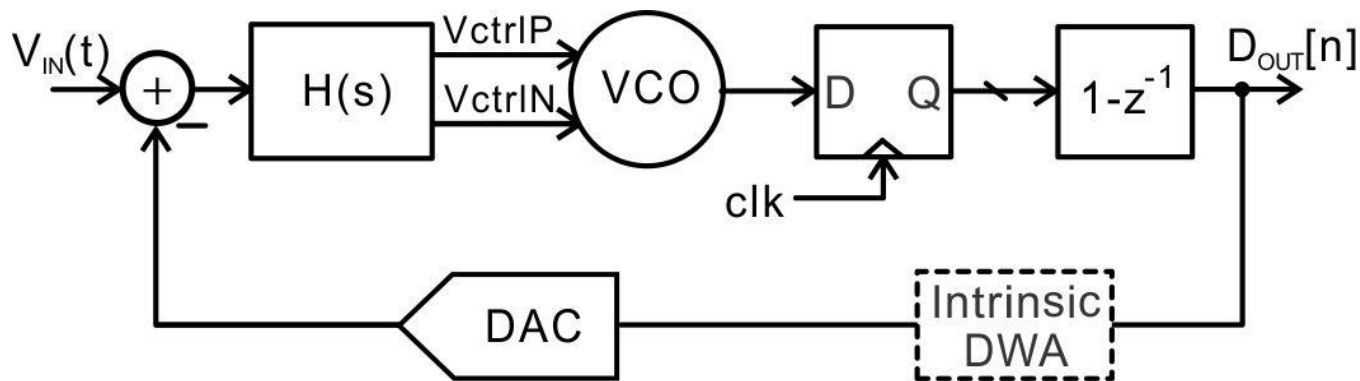


- Need conventional integrators (power and area)
- Have intrinsic 1st-order mismatch noise shaping
- VCO frequency is used as the output variable → large swing at the VCO input → large nonlinearity

[M. Z. Straayer and M. H. Perrott, IEEE JSSC, vol 43, no 4, pp. 805-814, April 2008]



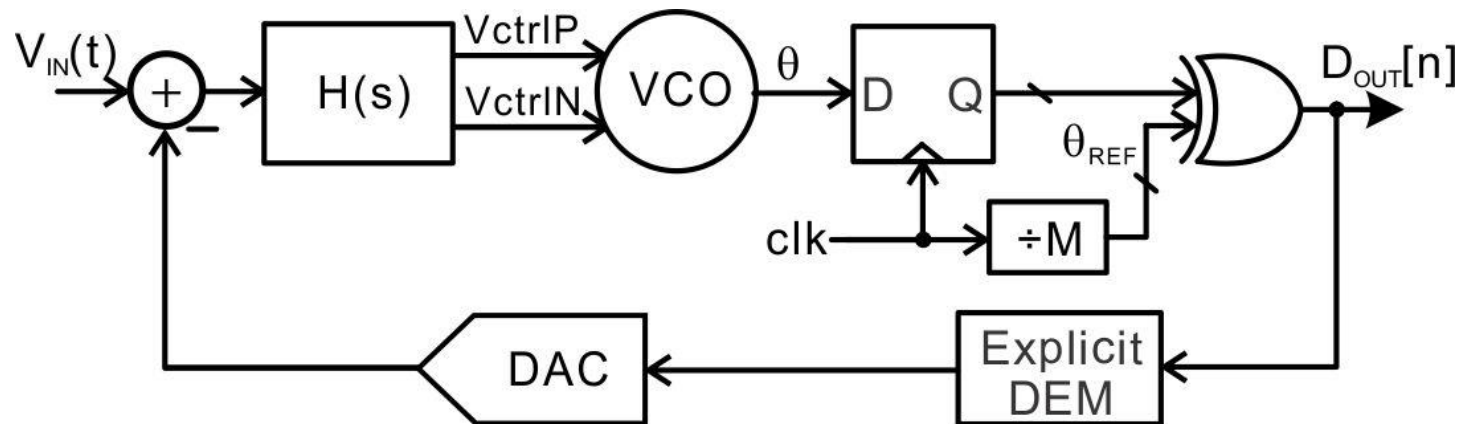
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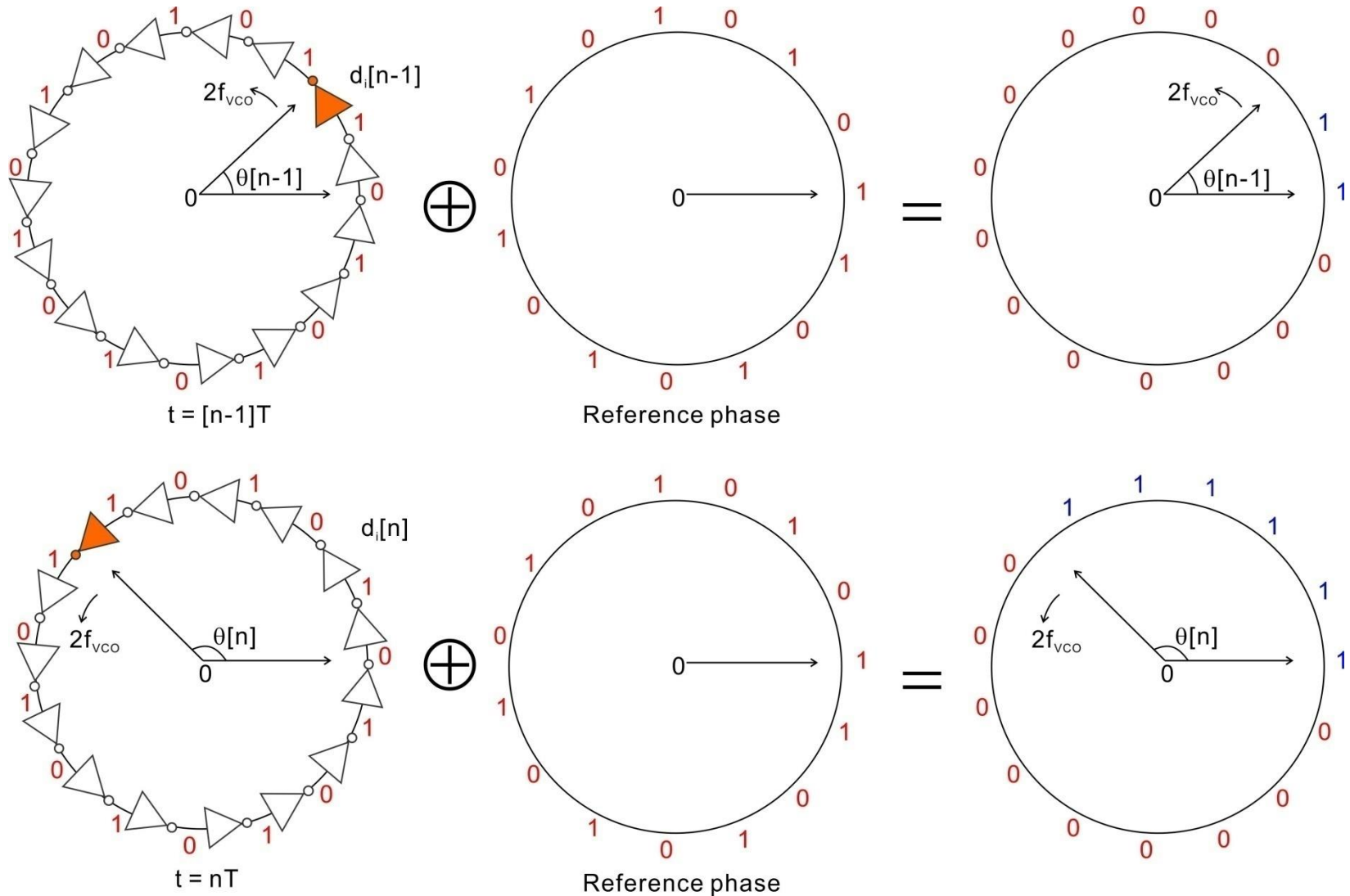
[M. Z. Straayer and M. H. Perrott, IEEE JSSC, vol 43, no 4, pp. 805-814, April 2008]

VCO-based ADC review (3)

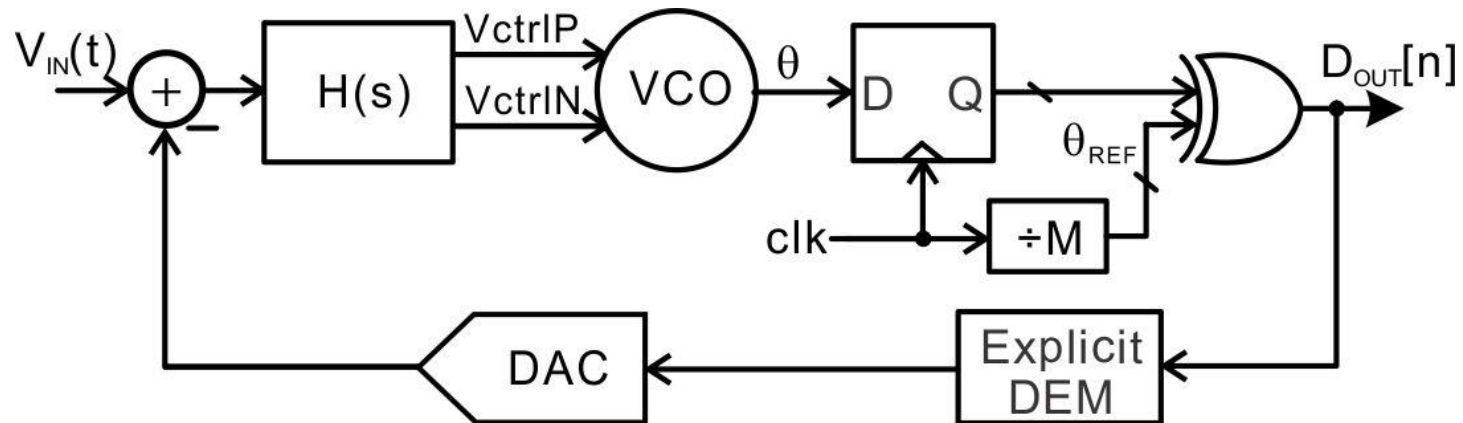


- VCO phase is used as the output variable $\rightarrow$ small swing at the VCO input $\rightarrow$ low nonlinearity
- Explicit DEM technique has to be used (delay)
- VCO frequency locked to sampling frequency

VCO behavior for $M=2$ ($f_s=2f_{vco}$)

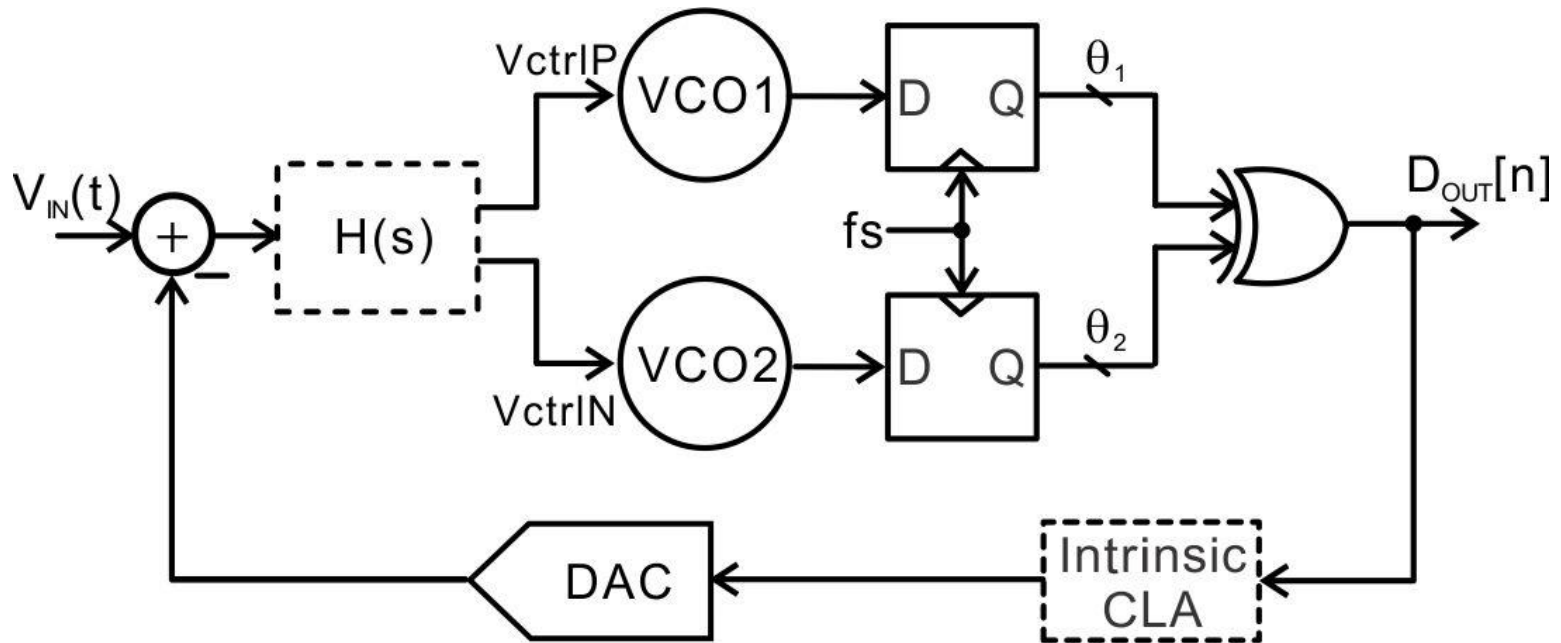


VCO-based ADC review (3)



- VCO phase is used as the output variable $\rightarrow$ small swing at the VCO input $\rightarrow$ low nonlinearity
- Explicit DEM technique has to be used (delay)
- VCO frequency locked to sampling frequency (like a PLL)

Proposed VCO-based $\Delta\Sigma$ ADC



- Decouple f_{VCO} from f_s (low f_{VCO} reduces VCO power)
- Even-order distortions are cancelled
- Intrinsic clocked averaging (CLA)

Intrinsic CLA behavior

